Cell Tuition
1-1 PRIVATE TUTORS

GEOMETRY

Map Scales and Directions

Name _____ Date _____ No calculator Version 1

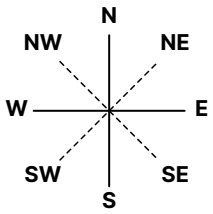
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KEY WORDS compass point · clockwise · scale *(tauine)* · grid reference

A Compass points and turns ●○○

> **Remember:** There are 45° between one compass point and the next. A quarter turn is 90°, a half turn is 180° and a full turn is 360°.

1. Which way do you face after each turn?



a) You face north and turn 90° clockwise. _____

b) You face east and turn 180° clockwise. _____

c) You face south and turn 90° anticlockwise. _____

d) You face north and turn 45° clockwise. _____

e) You face west and turn 135° clockwise. _____

f) You face south-east and turn 90° anticlockwise. _____

2. What angle do you turn through, clockwise?

a) From north to east _____ **b)** From north to south-east _____

c) From east to west _____ **d)** From south to north-west _____

e) From north-east to south _____ **f)** From west to south _____

3. Write the direction that is opposite to each of these.

a) North _____ **b)** South-east _____ **c)** West _____ **d)** North-east _____

B Map scales ●●○

> **Remember:** A scale tells you what 1 cm on the map stands for. If 1 cm represents 5 km, then 4 cm represents $4 \times 5 = 20$ km.

4. Find the real distance.

a) 1 cm represents 5 km. A distance on the map is 4 cm.

b) 1 cm represents 2 km. A distance on the map is 7.5 cm.

c) 1 cm represents 50 m. A distance on the map is 6 cm.

d) 1 cm represents 20 km. A distance on the map is 3.5 cm.

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5. Find the distance on the map.

a) 1 cm represents 5 km. A real distance is 35 km.

b) 1 cm represents 2 km. A real distance is 13 km.

c) 1 cm represents 50 m. A real distance is 450 m.

d) 1 cm represents 20 km. A real distance is 150 km.

C Scales written as ratios ●●●

⚠ **Watch out:** A scale of 1 : 50,000 means 1 cm on the map is 50,000 cm in real life. Change the answer to metres (÷ 100) or kilometres (÷ 100,000).

6. Answer each question.

a) The scale is 1 : 1,000. A distance on the map is 6 cm. Find the real distance in metres.

b) The scale is 1 : 50,000. A distance on the map is 3 cm. Find the real distance in kilometres.

c) The scale is 1 : 200. A distance on the map is 4.5 cm. Find the real distance in metres.

d) The scale is 1 : 100,000. A distance on the map is 7.5 cm. Find the real distance in kilometres.

D Grid references ●●○

💡 **Remember:** A grid reference names a square: the letter first (across), then the number (up). North is at the top of the map.

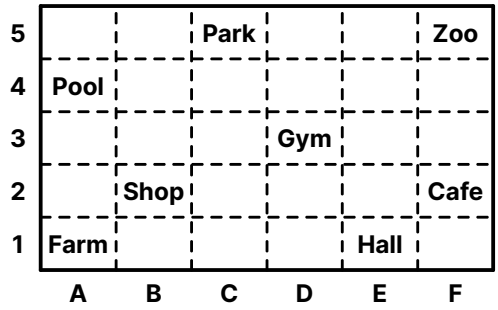
7. Use the map. Write the grid reference of each place.

a) The gym _____

b) The pool _____

c) The zoo _____

d) The farm _____



8. Which place is in each square?

a) Square C5 _____ **b)** Square B2 _____

9. Follow each set of directions on the map.

a) Start at the shop. Go 2 squares east and then 1 square north. Where are you? _____

b) Start at the zoo. Go 3 squares west. Where are you? _____

c) Start at the farm. Go 3 squares north. Where are you? _____

d) Start at the cafe. Go 1 square west and then 1 square south. Where are you? _____

★ Extension

10. The scale is 1 : 25,000. A distance on the map is 8 cm. Find the real distance in kilometres.

11. A walking track is 14 km long. How long is it on a map with a scale of 1 : 50,000?

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Map Scales and Directions

Name _____ Date _____ No calculator Version 2

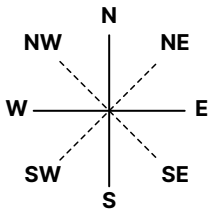
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KEY WORDS compass point · clockwise · scale *(tauine)* · grid reference

A Compass points and turns ●○○

> **Remember:** There are 45° between one compass point and the next. A quarter turn is 90°, a half turn is 180° and a full turn is 360°.

1. Which way do you face after each turn?



a) You face north and turn 90° anticlockwise. _____

b) You face west and turn 180° clockwise. _____

c) You face east and turn 90° clockwise. _____

d) You face north and turn 45° anticlockwise. _____

e) You face south and turn 135° clockwise. _____

f) You face north-east and turn 90° clockwise. _____

2. What angle do you turn through, clockwise?

a) From north to south _____ **b)** From north to north-east _____

c) From east to south _____ **d)** From west to north-east _____

e) From south-east to west _____ **f)** From south to east _____

3. Write the direction that is opposite to each of these.

a) South _____ **b)** South-west _____ **c)** East _____ **d)** North-west _____

B Map scales ●●○

> **Remember:** A scale tells you what 1 cm on the map stands for. If 1 cm represents 5 km, then 4 cm represents $4 \times 5 = 20$ km.

4. Find the real distance.

a) 1 cm represents 4 km. A distance on the map is 6 cm.

b) 1 cm represents 2 km. A distance on the map is 8.5 cm.

c) 1 cm represents 50 m. A distance on the map is 8 cm.

d) 1 cm represents 20 km. A distance on the map is 4.5 cm.

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5. Find the distance on the map.

a) 1 cm represents 4 km. A real distance is 36 km.

b) 1 cm represents 2 km. A real distance is 15 km.

c) 1 cm represents 50 m. A real distance is 350 m.

d) 1 cm represents 20 km. A real distance is 130 km.

C Scales written as ratios ●●●

⚠ **Watch out:** A scale of 1 : 50,000 means 1 cm on the map is 50,000 cm in real life. Change the answer to metres (÷ 100) or kilometres (÷ 100,000).

6. Answer each question.

a) The scale is 1 : 1,000. A distance on the map is 8 cm. Find the real distance in metres.

b) The scale is 1 : 50,000. A distance on the map is 5 cm. Find the real distance in kilometres.

c) The scale is 1 : 200. A distance on the map is 3.5 cm. Find the real distance in metres.

d) The scale is 1 : 100,000. A distance on the map is 4.5 cm. Find the real distance in kilometres.

D Grid references ●●○

Remember: A grid reference names a square: the letter first (across), then the number (up). North is at the top of the map.

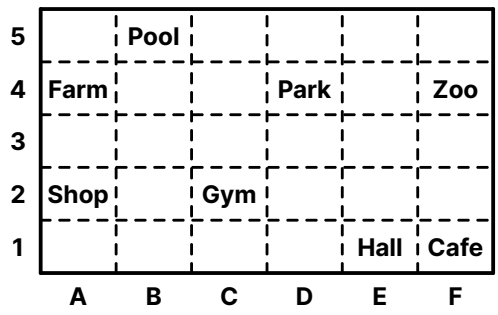
7. Use the map. Write the grid reference of each place.

a) The gym _____

b) The pool _____

c) The zoo _____

d) The farm _____



8. Which place is in each square?

a) Square D4 _____ **b)** Square A2 _____

9. Follow each set of directions on the map.

a) Start at the shop. Go 2 squares east. Where are you? _____

b) Start at the zoo. Go 2 squares west. Where are you? _____

c) Start at the hall. Go 1 square east. Where are you? _____

d) Start at the farm. Go 1 square east and then 1 square north. Where are you? _____

★ Extension

10. The scale is 1 : 25,000. A distance on the map is 12 cm. Find the real distance in kilometres.

11. A walking track is 9 km long. How long is it on a map with a scale of 1 : 50,000?

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